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b)

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -1 - 1 = -2 \neq 0 \Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{bmatrix} \text{ is invertible} \Rightarrow \{(1, 1, -1), (1, -1, 1), (0, 0, 1)\} \text{ is a basis of } \mathbb{R}^3.$$

e)

$$\begin{vmatrix} 2 & 1 & 0 & -1 \\ 1 & 1 & 1 & 2 \\ -1 & 0 & 0 & 3 \\ 3 & 2 & -3 & 1 \end{vmatrix} \xrightarrow{C_4 + 3C_1} \begin{vmatrix} 2 & 1 & 0 & 5 \\ 1 & 1 & 1 & 5 \\ -1 & 0 & 0 & 0 \\ 3 & 2 & -3 & 10 \end{vmatrix} \xrightarrow{C_2 \leftrightarrow C_1} - \begin{vmatrix} 1 & 0 & 5 \\ 1 & 1 & 5 \\ 2 & -3 & 10 \end{vmatrix} \xrightarrow{C_2 - C_1} \begin{vmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \\ 2 & -3 & 0 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 2 & 1 & 0 & -1 \\ 1 & 1 & 1 & 2 \\ -1 & 0 & 0 & 3 \\ 3 & 2 & -3 & 1 \end{vmatrix} \text{ is not invertible}$$

$$\Rightarrow \{(2, 1, -1, 3), (1, 1, 0, 2), (0, 1, 0, -3), (-1, 2, 3, 1)\} \text{ is not a basis of } \mathbb{R}^4.$$

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a) FALSE: $\{\vec{x}, \vec{y}, \vec{x} + \vec{y}\}$ is dependent because $1 \cdot \vec{x} + 1 \cdot \vec{y} + (-1)(\vec{x} + \vec{y}) = \vec{0}$.

c) TRUE: Because $\{\vec{y}, \vec{z}\}$ is dependent, there is a linear combination $a\vec{y} + b\vec{z} = \vec{0}$ where either $a=0$ or $b=0$. Thus, $\{\vec{x}, \vec{y}, \vec{z}\}$ is dependent because $0 \cdot \vec{x} + a\vec{y} + b\vec{z} = \vec{0}$.

f) FALSE: Take for instance $\vec{x} = (1, 1, 1)$, $\vec{y} = (-1, -1, -1)$, $\vec{z} = (0, 0, 0)$. Then $\vec{x} + \vec{y} + \vec{z} = \vec{0}$ and $\{\vec{x}, \vec{y}, \vec{z}\}$ is dependent or null.

k) TRUE: Let $\{\vec{x}_1, \dots, \vec{x}_k\}$ be a spanning set and $\{\vec{x}_1, \dots, \vec{x}_m, \vec{y}_1, \dots, \vec{y}_s\}$ a set containing it. If $\vec{v} \in \mathbb{R}^n$ is any vector, there exist $a_1, \dots, a_k \in \mathbb{R}$ with

$$\vec{v} = a_1 \vec{x}_1 + \dots + a_k \vec{x}_k = a_1 \vec{x}_1 + \dots + a_k \vec{x}_k + 0 \vec{y}_1 + \dots + 0 \vec{y}_s \in \text{span}\{\vec{x}_1, \dots, \vec{x}_k, \vec{y}_1, \dots, \vec{y}_s\}$$

Therefore, $\mathbb{R}^n = \text{span}\{\vec{x}_1, \dots, \vec{x}_k, \vec{y}_1, \dots, \vec{y}_s\}$.

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Suppose $a, b, c \in \mathbb{R}$ are such that $a\vec{x}_1 + b\vec{x}_3 + c\vec{x}_5 = \vec{0}$. Then we have $0\vec{x}_1 + a\vec{x}_2 + b\vec{x}_3 + 0\vec{x}_4 + c\vec{x}_5 + 0\vec{x}_6 = \vec{0}$. Since $\{\vec{x}_1, \dots, \vec{x}_6\}$ is independent, all the coefficients of the previous linear combination must be zero, so $a=b=c=0$. Thus, $\{\vec{x}_1, \vec{x}_3, \vec{x}_5\}$ is independent.

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Since $\dim W = 1$, W has a basis $B = \{\vec{x}\}$. Let $U \subseteq W$ be a subspace. Note that $\dim U \leq \dim W$.

• If $U = \{\vec{0}\}$, then we are done.

• If $U \neq \{\vec{0}\}$, then $\begin{cases} U \subseteq W \\ \dim U = \dim W = 1 \end{cases} \Rightarrow U = W$.

Therefore, $U = \{\vec{0}\}$ or $U = W$.

SECTION 5.4

1

a) The reduced row-echelon form of A is

$$\begin{bmatrix} 2 & -4 & 6 & 8 \\ 2 & -1 & 3 & 2 \\ 4 & -5 & 9 & 10 \\ 0 & -1 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

↑↑
Leading 1's

Therefore:

• Basis of row A : $\{(1, 0, 1, 0), (0, 1, -1, -2)\}$

• $\text{rank}(A) = 2$

• Basis of col A : $\{(2, 2, 4, 0), (-4, -1, -5, -1)\}$

b) The reduced row-echelon form of A is

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ -3 & -6 & 3 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

↑↑
Leading 1's

We got:

• Basis of row A : $\{(1, 2, -1, 3), (0, 0, 0, 1)\}$

• $\text{rank } A = 2$

• Basis of col A : $\{(1, 0), (0, 1)\}$

2

a) $U = \text{row } A$ for

$$A = \begin{bmatrix} 1 & -1 & 0 & 3 \\ 2 & 1 & 5 & 1 \\ 4 & -2 & 5 & 7 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

↑↑↑
Leading 1's

Thus, a basis for U is

$$\{(1, 0, 0, 3), (0, 1, 0, 0), (0, 0, 1, -1)\}$$

d) $U = \text{row } A$ for

$$A = \begin{bmatrix} 1 & 5 & -6 \\ 2 & 6 & -8 \\ 3 & 7 & -10 \\ 4 & 8 & 12 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, a basis for U is

$$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$